HOPPER INFORMATION SERIES

APRILTAGS

AprilTags with Hopper

Use printed tag36h11 landmarks for identity, relative image position, orientation, and bounded centering.

TAG36H11 | IDS 0–586 | RELATIVE POSE

AprilTags were built to give robots reliable visual landmarks

2011
AprilTag introduced

Edwin Olson, University of Michigan APRIL Lab; ICRA paper.

2016
AprilTag 2

A redesigned detector improved speed, detection rate, and false-positive behavior.

AprilTag 3
Faster + flexible

Small-tag detection, pose estimation, and more layouts/families.

TODAY
Robotics standard

Open implementations in research, ROS, calibration, navigation, and labs.

The original goal: print an artificial feature that a camera can identify and localize—even when natural visual features are unreliable.

A printed tag can become a shared coordinate reference

LOCALIZATION

- Estimate a robot or camera pose relative to a known landmark.

NAVIGATION + DOCKING

- Center on a landing pad, charging station, doorway, or work cell.

CALIBRATION

- Relate cameras, robot frames, and measurement rigs with known geometry.

MANIPULATION

- Identify bins, tools, parts, or target objects for a robot arm.

SLAM + GROUND TRUTH

- Supply repeatable landmarks while testing perception and state estimation.

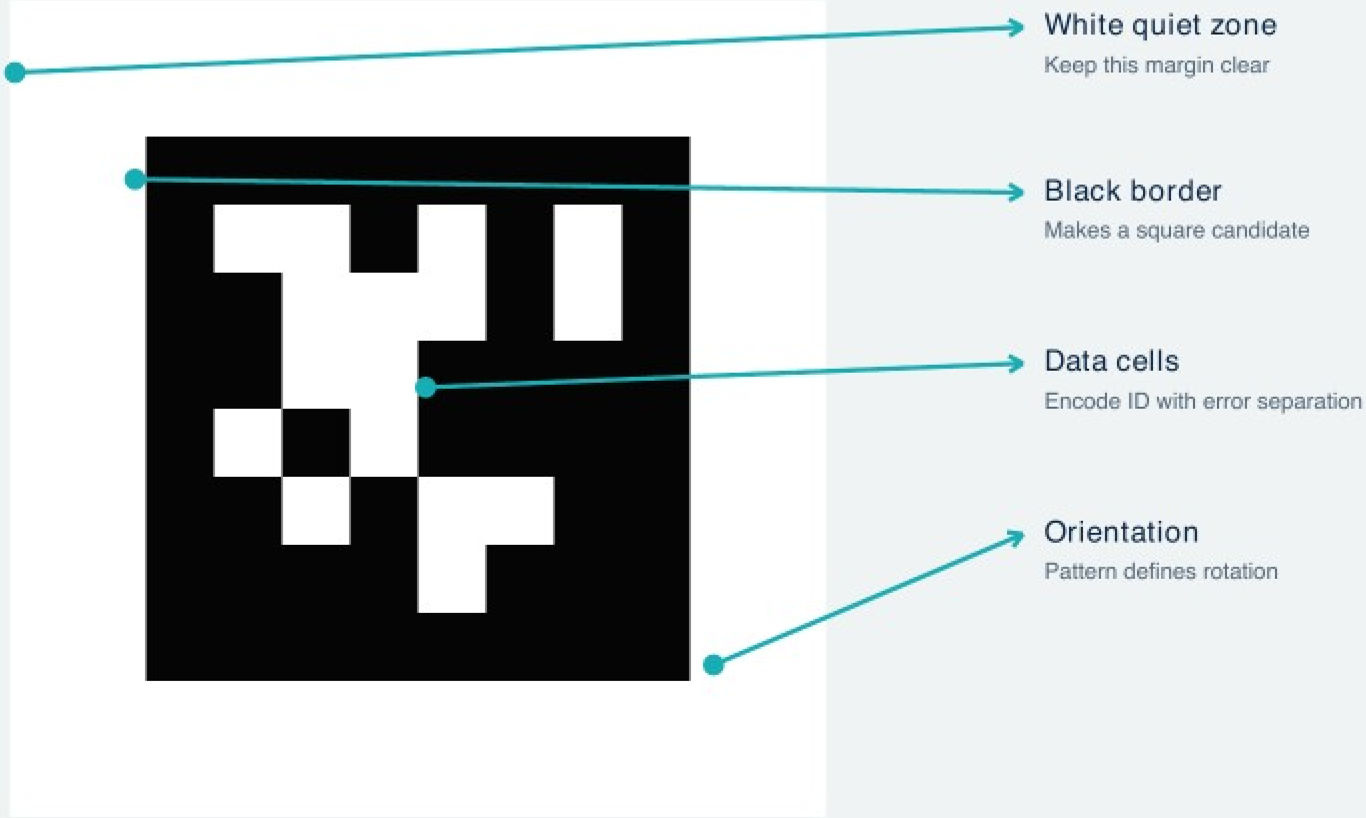
AR + HUMAN INTERFACES

- Anchor digital information to a physical place or object.

A tag is useful because its identity and geometry are designed—not discovered by chance.

TAG ANATOMY

A visual ID designed for robots



tag36h11 ID 0

HOPPER FAMILY

- tag36h11
- IDs 0 through 586
- Black and white only
- Up to 3 payload-bit errors accepted

WHY FIDUCIALS

- Known geometry.
- Unique machine-readable ID.
- Orientation from one image.
- No battery or radio on the tag.

Family chooses the codebook; ID chooses one tag inside it

FAMILY	CODE DESIGN	WHEN IT FITS
tag16h5	Small code, fewer IDs	Legacy / compact demonstrations
tag25h9	More bits and separation	Legacy balance of size and robustness
tag36h11	36 data bits · minimum Hamming distance 11	Hopper Studio: IDs 0–586
tagStandard41h12	Modern standard layout	Upstream default for most new work
tagCircle49h12	Circular, space-efficient layout	Small circular objects

Never mix families in one detector configuration: the same ID number in another family is a different code.

LOCAL TAG36H11 PIPELINE

How Hopper Studio detects a tag

1 **Grayscale**
Convert RGB camera pixels to luminance.

→

2 **Threshold**
Estimate a black/white split.

→

3 **Squares**
Find connected dark square candidates.

→

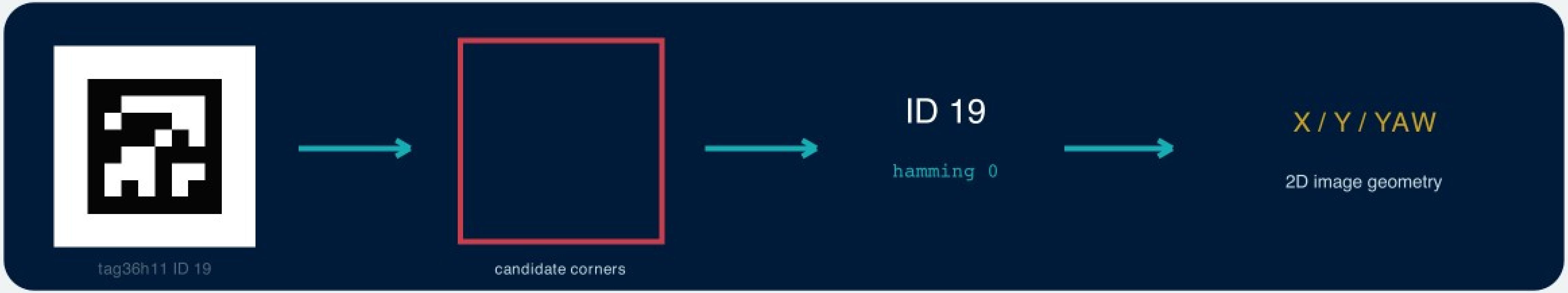
4 **Sample**
Map each candidate into an 8x8 grid.

→

5 **Match**
Compare rotations with the codebook.

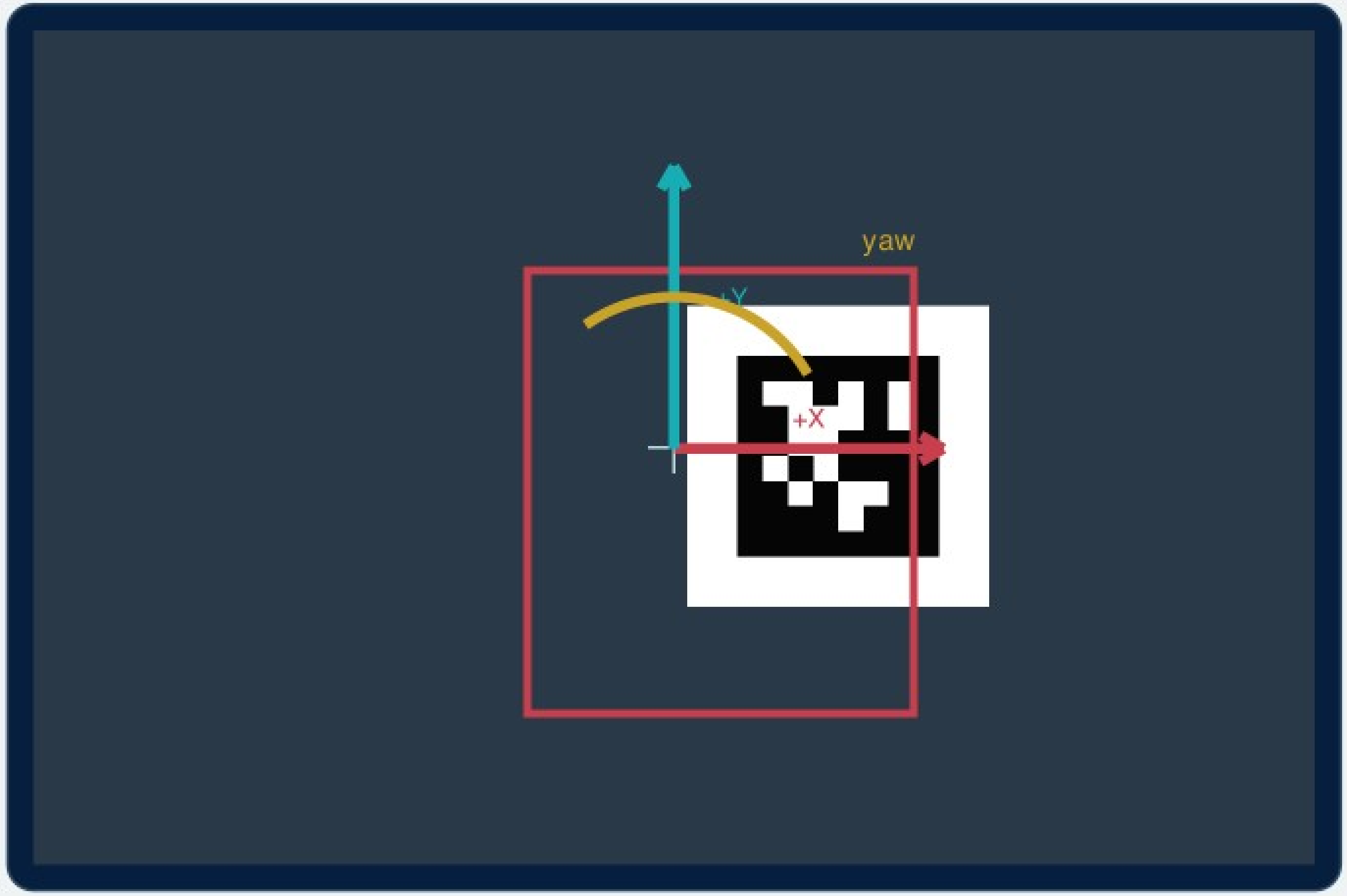
→

6 **Pose**
Return ID, corners, center and 2D yaw.



WHAT THE RESULT MEANS

Coordinates and pose in Hopper Studio



APRILTAG RESULT

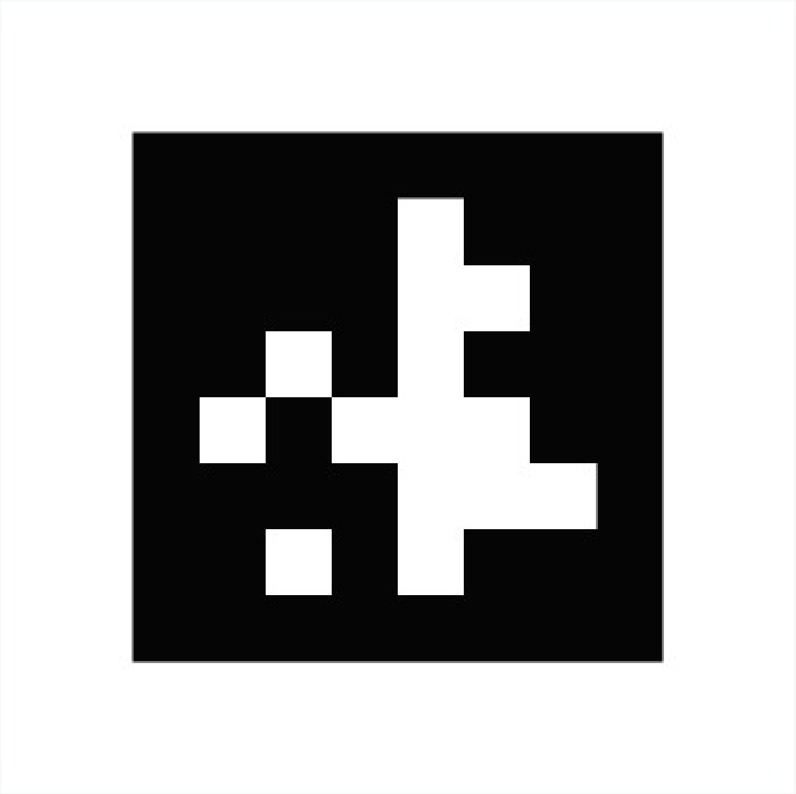
- id + family + hamming
- four corners + center in pixels
- bounding box
- centerX / centerY: -100..100
- yaw: normalized around +/-180 deg

DO NOT OVERCLAIM

- Yaw is 2D image-plane orientation.
- It is not full 3D pose or aircraft heading.
- No distance or altitude is returned.
- centerOnAprilTag never lands.

PHYSICAL TAG SETUP

Print cleanly and place deliberately



tag36h11 ID 7

PRINT IN HOPPER STUDIO

- Vision Testing -> AprilTag Detection.
- Choose an ID from 0-586.
- Generate PDF for a full-page US Letter vector tag.
- Print at 100% scale with a clean white margin.

PLACE FOR THE CAMERA

- Keep the sheet flat and well lit.
- Avoid glare, folds, shadows and clipped margins.
- Start large and close; test before increasing height.
- Keep paper and people clear of propellers.

Centering needs tolerances, timeouts, and landing

BOUNDED APRILTAG CENTERING · PROGRAM.JS

```
await drone.takeOff();
try {
  const centered = await vision.centerOnAprilTag(
    drone,
    7,
    10,
    5,
    5,
    3,
  );

  if (!centered) {
    console.warn("Tag 7 was not centered.");
  }
} finally {
  await drone.land();
}
```

CENTER AND ALIGN

```
await vision.centerOnAprilTag(
  drone, id, power, centerSlack, angleSlack, lostSearches)
```

drone — DroneController used for corrective motion
id — 0…586 or "any"; default "any"
power — translation 0…100%; default 10
centerSlack — 1…35% of frame; default 5
angleSlack — 1…45 degrees; default 5
lostSearches — 1…20 scans; default 3
returns — Promise<boolean>; also has a 30 s timeout
does not — control altitude or land the aircraft

SCAN ALL TAGS

```
await vision.scanAprilTags(announceScan)
```

announceScan: boolean; default true · returns Promise<AprilTagDetection[]>
fields: id, bbox, corners, centerX/Y, yaw, hamming · throws on camera failure

TEST ONE ID

```
await vision.seesAprilTag(id)
```

id: 0…586 or "any"; default "any" · numeric IDs are rounded
returns Promise<boolean> after a fresh scan · throws on camera failure

CHECKLIST AND SOURCES

Reliable tags need reliable experiments

BEFORE FLIGHT

- Confirm the family is tag36h11.
- Test the intended ID, size, angle and lighting.
- Keep the full border and quiet zone in frame.
- Use a lost-tag limit and an overall timeout.
- Plan a safe fallback land.

SOURCES AND FURTHER READING

University of Michigan APRIL Lab - AprilTag
https://april.eecs.umich.edu/software/apriltag

AprilTag 2 paper
https://april.eecs.umich.edu/media/pdfs/wang2016iros.pdf

Hopper Studio detector and printable tag generator
lib/apriltags.ts, lib/vision.ts and README.md

DETECT -> CENTER -> ALIGN -> DECIDE

The mission still owns height, obstacle clearance and landing.